

COUNTEREXAMPLES TO LEW'S H -FREE LAPLACIAN CONJECTURE

ABSTRACT. For a graph G , let

$$\varepsilon_k(G) = \sum_{i=1}^k \lambda_i(L(G)) - |E(G)|,$$

where the Laplacian eigenvalues are ordered nonincreasingly. Lew conjectured that every H -free graph G with $|V(G)| \geq k$ satisfies $\varepsilon_k(G) \leq \text{ex}(k+1; H)$. We disprove the conjecture as stated: the path P_4 is triangle-free, but

$$\varepsilon_2(P_4) = 1 + \sqrt{2} > 2 = \text{ex}(3; K_3).$$

This example is minimal in both k and $|V(G)|$ when the forbidden graph has an edge and no isolated vertices. More generally, every graph H with $\delta(H) \geq 2$ admits a connected H -free counterexample, and for every $r \geq 3$ and $k \geq r-1$ there is a connected K_r -free counterexample at parameter k .

1. THE SMALLEST COUNTEREXAMPLE

All graphs are finite and simple, and H -free means containing no copy of H as a subgraph. Let

$$\lambda_1(L(G)) \geq \cdots \geq \lambda_{|V(G)|}(L(G)) = 0$$

be the Laplacian eigenvalues of G , and let $\text{ex}(n; H)$ be the maximum number of edges in an n -vertex H -free graph. Conjecture 7.1 of Lew [1] asserts that

$$(1) \quad \varepsilon_k(G) \leq \text{ex}(k+1; H)$$

for every $k \geq 1$, every graph H , and every H -free graph G with $|V(G)| \geq k$. Verbatim, Conjecture 7.1 of [1] reads: “Let $k \geq 1$, and let H be a graph. Then, for every H -free graph $G = (V, E)$ with $|V| \geq k$, $\varepsilon_k(G) \leq \text{ex}(k+1; H)$.” It is this statement, with no restriction on H beyond being a graph, that we refute.

We write P_j for the path on j vertices.

Theorem 1. *The inequality (1) is false. Specifically, for $k = 2$, $H = K_3$, and $G = P_4$,*

$$\varepsilon_2(G) = 1 + \sqrt{2} > 2 = \text{ex}(3; K_3).$$

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Proof. A direct calculation gives

$$\det(tI - L(P_4)) = t(t-2)(t^2 - 4t + 2),$$

so the Laplacian eigenvalues of P_4 are

$$2 + \sqrt{2}, \quad 2, \quad 2 - \sqrt{2}, \quad 0.$$

Since P_4 has three edges,

$$\varepsilon_2(P_4) = (2 + \sqrt{2}) + 2 - 3 = 1 + \sqrt{2}.$$

The graph P_4 is triangle-free, while Mantel's theorem gives $\text{ex}(3; K_3) = 2$. $\square$

2. PENDANT AMPLIFICATION

For a graph F and an integer $N \geq 1$, let $F^{(N)}$ be the graph obtained by attaching N new pendant vertices to each vertex of F .

Lemma 2. *Let F be a connected graph with $k \geq 2$ vertices, e edges, and Laplacian eigenvalues $\mu_1, \dots, \mu_k$. Put $G = F^{(N)}$. Then the Laplacian spectrum of G consists of 1 with multiplicity $k(N-1)$ and, for each i , two numbers α_i, β_i that are the roots of*

$$(2) \quad t^2 - (N+1+\mu_i)t + \mu_i = 0$$

ordered so that $0 \leq \alpha_i < 1 < \beta_i$. Moreover,

$$(3) \quad \varepsilon_k(G) = k + e - \sum_{i=1}^k \alpha_i > k + e - \frac{2e}{N+1}.$$

Proof. For each vertex of F , the vectors supported on its N pendant neighbors and having coordinate sum zero give the eigenvalue 1. Their total dimension is $k(N-1)$.

Let x be a Laplacian eigenvector of F with eigenvalue μ . On vectors whose core coordinates are ax and whose pendant coordinates adjacent to the i th core vertex are all bx_i , the Laplacian acts on (a, b) through

$$\begin{pmatrix} \mu + N & -N \\ -1 & 1 \end{pmatrix}.$$

Its characteristic polynomial is (2). The pendant-difference vectors of the previous paragraph vanish on the core and have coordinate sum zero on each pendant block, whereas the vectors just described are constant on each pendant block; hence the two families are orthogonal, and in particular linearly independent. Since $k(N-1)+2k = k(N+1) = |V(G)|$, together they form a basis of $\mathbb{R}^{V(G)}$, so the numbers listed above are the whole Laplacian spectrum of G .

For $\mu > 0$, the polynomial in (2) is positive at 0 and negative at both 1 and $N+1$. Hence its smaller root lies in $(0, 1)$ and its larger root exceeds $N+1$. For $\mu = 0$, the roots are 0 and $N+1$. Thus the k largest Laplacian eigenvalues of G are the β_i .

Since $\alpha_i + \beta_i = N + 1 + \mu_i$ and $\sum_i \mu_i = 2e$,

$$\sum_{i=1}^k \beta_i = k(N + 1) + 2e - \sum_{i=1}^k \alpha_i.$$

The graph G has $kN + e$ edges, which gives the identity in (3). Finally, $\alpha_i \beta_i = \mu_i$ and $\beta_i > N + 1$ whenever $\mu_i > 0$, so

$$\sum_{i=1}^k \alpha_i < \frac{\sum_i \mu_i}{N + 1} = \frac{2e}{N + 1}.$$

□

Proposition 3. *Let H satisfy $\delta(H) \geq 2$, and let F be a connected H -free graph with $k \geq 2$ vertices and e edges. Set*

$$d = k + e - \text{ex}(k + 1; H).$$

If $d > 0$ and $2e/(N + 1) \leq d$, then $F^{(N)}$ is a connected H -free counterexample to (1) at parameter k .

Proof. A copy of H cannot use a pendant vertex, because every vertex of H has degree at least two. Hence $F^{(N)}$ is H -free, and it is connected because F is connected. Lemma 2 gives

$$\varepsilon_k(F^{(N)}) > k + e - \frac{2e}{N + 1} \geq k + e - d = \text{ex}(k + 1; H).$$

□

Corollary 4. *For every graph H with $\delta(H) \geq 2$, there is a connected H -free counterexample to (1).*

Proof. Let $h = |V(H)|$, set $k = h - 1$, and take $F = K_k$. Since $\delta(H) \geq 2$ while degrees in a simple graph are at most $h - 1$, we have $h \geq 3$; hence $k = h - 1 \geq 2$ and $e = \binom{k}{2} \geq 1$, as required by Lemma 2 and Proposition 3 and so that $N = 2e \geq 1$. Then F is connected and H -free. Moreover,

$$k + e = \binom{h}{2} \quad \text{and} \quad \text{ex}(h; H) \leq \binom{h}{2} - 1,$$

because K_h contains H . Thus $d \geq 1$. Since $2e/(2e + 1) < 1$, taking $N = 2e$ in Proposition 3 proves the claim. □

Corollary 5. *For every $r \geq 3$ and every $k \geq r - 1$, there is a connected K_r -free graph G such that*

$$\varepsilon_k(G) > \text{ex}(k + 1; K_r).$$

Proof. Let $F = T_{r-1}(k)$ be the balanced complete $(r - 1)$ -partite graph on k vertices, and write

$$q = |E(F)| = \text{ex}(k; K_r).$$

By Turán's theorem, adding one vertex to a smallest part gives

$$\text{ex}(k+1; K_r) - \text{ex}(k; K_r) = k - \left\lfloor \frac{k}{r-1} \right\rfloor.$$

Consequently,

$$k + q - \text{ex}(k+1; K_r) = \left\lfloor \frac{k}{r-1} \right\rfloor \geq 1.$$

The graph F is connected because $k \geq r-1$. Since $2q/(2q+1) < 1$, Proposition 3, applied with $H = K_r$ and $N = 2q$, proves the claim. $\square$

Remark 6. The violation produced above is small. By the identity in (3), with α_i as in Lemma 2 and $F = T_{r-1}(k)$, the graph $G = T_{r-1}(k)^{(2q)}$ satisfies

$$\varepsilon_k(G) - \text{ex}(k+1; K_r) = \left\lfloor \frac{k}{r-1} \right\rfloor - \sum_{i=1}^k \alpha_i < \left\lfloor \frac{k}{r-1} \right\rfloor,$$

so the excess is $O(k)$, while $\text{ex}(k+1; K_r) = \Theta(k^2)$. The reason is visible in (3): pendant amplification gains less than k over $|E(F)|$, whereas passing from k to $k+1$ vertices raises the Turán number by $k - \lfloor k/(r-1) \rfloor$, falling short of k by exactly $\lfloor k/(r-1) \rfloor$. Thus it is the exact inequality (1) that fails, and not its asymptotic form: the counterexamples are consistent with the bound $\varepsilon_k(G) \leq \text{ex}(k; H) + (4k-2)\sqrt{k}$, valid for every H -free graph G and every $1 \leq k \leq |V(G)|$, recorded in [1, Section 7], and we do not know whether (1) holds up to a multiplicative $1+o(1)$ or an additive $o(k^2)$ error.

3. MINIMALITY

Proposition 7. *Suppose that H has at least one edge and no isolated vertices. Then (1) holds when $k = 1$, and it also holds whenever $|V(G)| \leq k+1$. Hence every counterexample under these assumptions satisfies*

$$k \geq 2, \quad |V(G)| \geq k+2 \geq 4.$$

The example in Theorem 1 attains both lower bounds.

Proof. First let $k = 1$. If G is edgeless, then $\varepsilon_1(G) = 0$. Otherwise choose a connected component C with $\lambda_1(L(C)) = \lambda_1(L(G))$, and put $t = |V(C)|$. Since C is connected,

$$t \leq |E(C)| + 1 \leq |E(G)| + 1.$$

Also $\lambda_1(L(C)) \leq t$: on the orthogonal complement of the all-ones vector,

$$L(C) + L(\overline{C}) = tI.$$

Therefore $\varepsilon_1(G) \leq 1$. If $H = K_2$, every H -free graph is edgeless. If $H \neq K_2$, then $\text{ex}(2; H) = 1$. Thus (1) holds for $k = 1$.

Now let $n = |V(G)| \leq k+1$. Since $n \geq k$, either $k = n$ or $k = n-1$. In both cases the sum of the k largest Laplacian eigenvalues is $2|E(G)|$, so

$$\varepsilon_k(G) = |E(G)|.$$

If $n = k + 1$, the definition of $\text{ex}(k + 1; H)$ gives the desired bound. If $n = k$, add one isolated vertex to G . Because H has no isolated vertices, the resulting $(k + 1)$ -vertex graph remains H -free, and the same bound follows. $\square$

Remark 8. The no-isolated-vertices hypothesis in Proposition 7 is necessary. For

$$H = K_2 \cup K_1, \quad G = K_2, \quad k = 2,$$

one has $\varepsilon_2(G) = 1 > 0 = \text{ex}(3; H)$. In particular, since (1) is asserted for every graph H , the conjecture in its literal form already fails for $G = K_2$ at $k = 2$, with only two vertices. The minimality assertion of Proposition 7 concerns forbidden graphs with an edge and no isolated vertices, and it is in that regime that P_4 is smallest.

REFERENCES

- [1] A. Lew, *An approximate version of Brouwer's Laplacian conjecture*, arXiv:2601.17575v1 [math.CO], 2026.